Everything In Moderation: The Currency-Maturity Composition of Sovereign Debt*

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September, 2026

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Abstract

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JEL Classification: E31, E43, E52, F34, H63 *Keywords:* currency choice, default, inflation, maturity, sovereign debt

*I am deeply indebted to Saroj Bhattarai, Olivier Coibion, Andrés Drenik, Oliver Pfäuti, Zachary Stangebye, Rachel Wellhausen, and Adrien Wicht for their invaluable guidance and feedback. I am grateful to Fernando Broner, Guido Lorenzoni, and Sergio Schmukler for sharing their data. I also thank Daniel Hsu, Hayden Parsley, and participants at the 2024 and 2026 Society for Computational Economics and the UT Austin Macro Writing Seminar for their helpful comments.

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Country	Sample Period	Number of Bonds	FC Share	Weighted Average Maturity
Angola	Nov. 2005 - Dec. 2024	2,344	9.8%	4.1 years
Argentina	Mar. 1991 - Dec. 2024	874	62.1%	10.9 years
Colombia	Dec. 1990 - Dec. 2024	763	11.8%	11.7 years
Egypt	Dec. 1993 - Dec. 2024	2,715	9.5%	2.0 years
Hungary	Jul. 1993 - Dec. 2024	2,246	13.8%	5.5 years
Indonesia	Jul. 1990 - Dec. 2024	1,146	14.5%	10.6 years
Jamaica	Sep. 1991 - Dec. 2024	778	55.5%	7.3 years
Jordan	Aug. 1993 - Nov. 2024	1,229	17.8%	4.3 years
Mexico	Mar. 1990 - Dec. 2024	2,450	7.2%	4.9 years
Philippines	May. 1990 - Dec. 2024	2,066	12.5%	6.5 years
Poland	Dec. 1993 - Nov. 2024	1,242	13.7%	6.4 years
Qatar	Feb. 1999 - Sep. 2024	238	55.0%	10.6 years
Romania	Jun. 1997 - Dec. 2024	888	38.0%	6.9 years
Russia	May. 1993 - May. 2024	571	12.4%	8.1 years
Tunisia	Sep. 1990 - Dec. 2024	281	21.7%	6.5 years
Türkiye	Mar. 1990 - Dec. 2024	992	15.8%	4.1 years
Ukraine	Mar. 1995 - Dec. 2022	1,813	42.8%	6.1 years
Uruguay	Jan. 1990 - Dec. 2024	348	52.2%	14.9 years
Venezuela	Dec. 1990 - Nov. 2018	1,926	35.0%	7.7 years

Table 1: Country-level issuance data description

I Introduction

The rest of the paper is as follows. I present novel empirical evidence on sovereigns' debt issuance choices in section II. The quantitative model is then outlined in section III. I discuss the economic forces that drive the sovereign's optimal debt portfolio in section IV. I calibrate my model to the Turkish economy in section V and show my model's predictions. section VII concludes.

II Empirical Analysis

In this section, I analyze data on sovereign bond issuance and spreads. Firstly, I robustly find that sovereign debt denominated in a country's domestic currency tends to be of significantly shorter maturity than debt denominated in a foreign currency. Secondly, I find that in times of economic distress, or high spreads, sovereigns tilt their issuance towards shorter maturities and towards more foreign currency debt.

Country	Sample Period	Number of Observations	Source
Colombia	Aug. 2003 - Dec. 2024	5,381	Du and Schreger (2016)
Egypt	Jan. 1997 - Oct. 2024	112	Datastream
Hungary	Jan. 2000 - Dec. 2024	6,450	Du and Schreger (2016)
Indonesia	Feb. 2003 - Dec. 2024	5,488	Du and Schreger (2016)
Jamaica	Jan. 1989 - Oct. 2024	144	Datastream
Mexico	Sep. 2002 - Dec. 2024	5,724	Du and Schreger (2016)
Philippines	Dec. 2000 - Dec. 2024	5,901	Du and Schreger (2016)
Poland	Jan. 2000 - Dec. 2024	6,452	Du and Schreger (2016)
Romania	Jan. 1994 - Oct. 2024	116	Datastream
Russia	Apr. 2006 - Feb. 2022	4,136	Du and Schreger (2016)
Türkiye	Feb. 2005 - Dec. 2024	5,107	Du and Schreger (2016)
Uruguay	Oct. 1991 - Oct. 2024	116	Datastream

Table 2: Country-level bond spread data description

II.A Data Description

I use three datasets in my empirical analysis. The first dataset is regarding sovereign bond issuance, which I collected from Bloomberg. I use data on issuance rather than stocks to analyze sovereigns' borrowing behvaior, because there is no publicly available data that is granular enough to decompose debt by currency denomination and maturity simultaneously.[1] The second and third datasets are from Datastream and Du and Schreger (2016), which contain data on sovereign bond spreads. Summary statistics are shown in Table 1 and Table 2.

II.A.1 Data on Issuance

My individual bond-level data from Bloomberg spans 19 emerging market economies from 1990 until 2024.[2] I include countries in my sample if (i) Bloomberg has at least 200 valid bond issuances for that country over the sample period, and (ii) at least 5% of the country's total amount of issued debt is denominated in a foreign currency. I impose a minimuim foreign currency threshold since the debt management offices of countries with very low foreign currency debt shares are unlikely to need to optimize their borrowing portfolio with respect to currency composition on top of maturity composition.[3]

There are two drawbacks of Bloomberg data. Firstly, it is imposible to know whether the

[1] Arslanalp and Tsuda (2012) provides quarterly frequency data on foreign holdings of sovereign debt but does not provide a breakdown by maturity.

[2] Bloomberg data only became available for most countries from 1990 onwards.

[3] This restriction excludes notable emerging market economies such as Brazil, China, India, Malaysia, and Lithuania. I omit Lithuania because it officially adopted the Euro as its national currency in 2015 and has since denominated all its debt in Euros. By amount issued, the foreign currency share of Brazil, China, India, and Malaysia's sovereign debt were only 2.9%, 1.50%, 0.19%, and 0.8%, respectively. Furthermore, India only started issuing non-rupee-denominated sovereign bonds in June 2019.

bond was bought by a domestic or international investor since there is no data on the creditor's nationality. To address this, I follow Gürkaynak et al. (2007) and Ottonello and Perez (2019) by removing all bonds issued by a country's central bank and bonds that mature in less than 3 months from issuance since they are typically only bought by domestic investors. Secondly, Bloomberg data also includes sub-national (or municipal) bonds. Since sub-national bonds have different risk profiles to national bonds,[4] I also remove them.[5]

II.A.2 Data on Spreads

I use panel data on sovereign bond spreads from Datastream and Du and Schreger (2016). For a given country, both datasets provide a time series of the yield differential between a bond issued by that country and denominated in their domestic currency, and the yield on a bond of the same maturity but issued by the US Treasury and denominated in USD. The resulting yield differential is the sovereign bond spread, which is a measure of the risk premium that investors demand to hold a country's debt relative to a risk-free benchmark.

The data from Du and Schreger (2016) is of daily frequency and covers eight emerging market economies that overlap with my issuance data. In order to expand the coverage across countries, I further use data from Datastream. Unfortunately, Datastream's frequency is only of monthly, quarterly, and annual frequency, and the monthly frequency data contains gaps in the time series. Therefore, I use Du and Schreger (2016)'s data where possible, and use quarterly frequency data from Datastream for the remaining four countries.

II.B Average Maturity by Currency

I report the average maturity of issued sovereign debt by currency denomination for each country in my empirical sample in Table 3. Columns 1 and 2 shows the weighted average of each country's domestic and foreign currency debt respectively, where each bond is weighted by the amount issued. Columns 3 and 4 show the unweighted average maturity.

Table 3 shows that across countries, the foreign currency debt that they issue is consistently of much longer maturity than the debt that is denominated in their own currency. This stylized fact holds across most countries, with Jamaica being the only exception. Across all bonds in my data, the weighted mean maturity of domestic currency debt is 4.7 years whereas it is over twice as long for foreign currency debt at 12.5 years.

Moreover, this empirical pattern is also consistent across time. I plot the time series of the weighted average maturities of domestic and foreign currency debt in Figure 1. Across every year in my 35-year sample, foreign currency debt has been of longer maturity than domestic currency debt.

Lastly, I present the distribution of maturities of domestic and foreign currency debt across all bonds in Figure 2. While foreign currency debt of shorter maturity (less than 3 years) and

[4] Schwert (2017) finds that municipal bonds pay an oversized risk premium for default risk relative to comparable bonds at the national level.

[5] I filter out all municipal bonds, bonds issued from to finance local projects, and bonds issued by state-owned companies. A bond is considered to be from a state-owned company if the issuer name contains strings such as 'Ltd.' or 'Corp.'

Country	Weighted		Unweighted	
	DC	FC	DC	FC
Angola	3.4 years	10.6 years	2.7 years	5.1 years
Argentina	6.5 years	13.5 years	3.5 years	8.3 years
Colombia	11.2 years	15.8 years	4.5 years	10.3 years
Egypt	1.6 years	6.6 years	1.5 years	5.6 years
Hungary	4.7 years	10.6 years	2.2 years	8.3 years
Indonesia	10.0 years	14.0 years	4.6 years	13.0 years
Jamaica	8.8 years	6.1 years	6.4 years	7.9 years
Jordan	3.7 years	6.8 years	3.3 years	7.0 years
Mexico	3.8 years	18.4 years	2.9 years	11.7 years
Philippines	5.2 years	15.1 years	2.5 years	11.8 years
Poland	5.6 years	11.6 years	2.3 years	11.5 years
Qatar	4.9 years	15.3 years	5.1 years	13.5 years
Romania	4.8 years	10.2 years	1.6 years	8.6 years
Russia	6.7 years	17.5 years	5.7 years	11.4 years
Tunisia	5.7 years	9.5 years	2.6 years	18.8 years
Türkiye	3.5 years	7.5 years	3.0 years	6.0 years
Ukraine	5.7 years	6.6 years	3.3 years	4.2 years
Uruguay	11.6 years	18.0 years	8.5 years	8.4 years
Venezuela	2.6 years	17.2 years	1.7 years	15.8 years
Average	5.8 years	12.2 years	3.6 years	9.9 years
Median	5.2 years	11.6 years	3.0 years	8.6 years
Standard Deviation	2.8 years	4.3 years	1.8 years	3.8 years

Notes: Country-level statistics (rows 1–19) are computed across bond issuances within their respective country. Moments (rows 20–22) are calculated across countries.

Table 3: Average Maturity of External Sovereign Debt by Currency

domestic currency debt of longer maturity (greater than 10 years) do get issued by countries, the overwhelming majority of long-term debt is denominated in foreign currency.

II.C Issuance Choices and Spreads

I now extend the analysis by analyzing the relationship between both margins of sovereign debt issuance and spreads. Due to the quarterly frequency of Datastream data and sovereigns issuing debt significantly less frequency than the daily frequency spread data from Du and Schreger (2016), all the following analysis is at the quarterly frequency unless otherwise specified.

The first column of Table 4 reports the correlation between spreads and the share of debt issued in a country's domestic currency for each quarter across each country. With the excep-

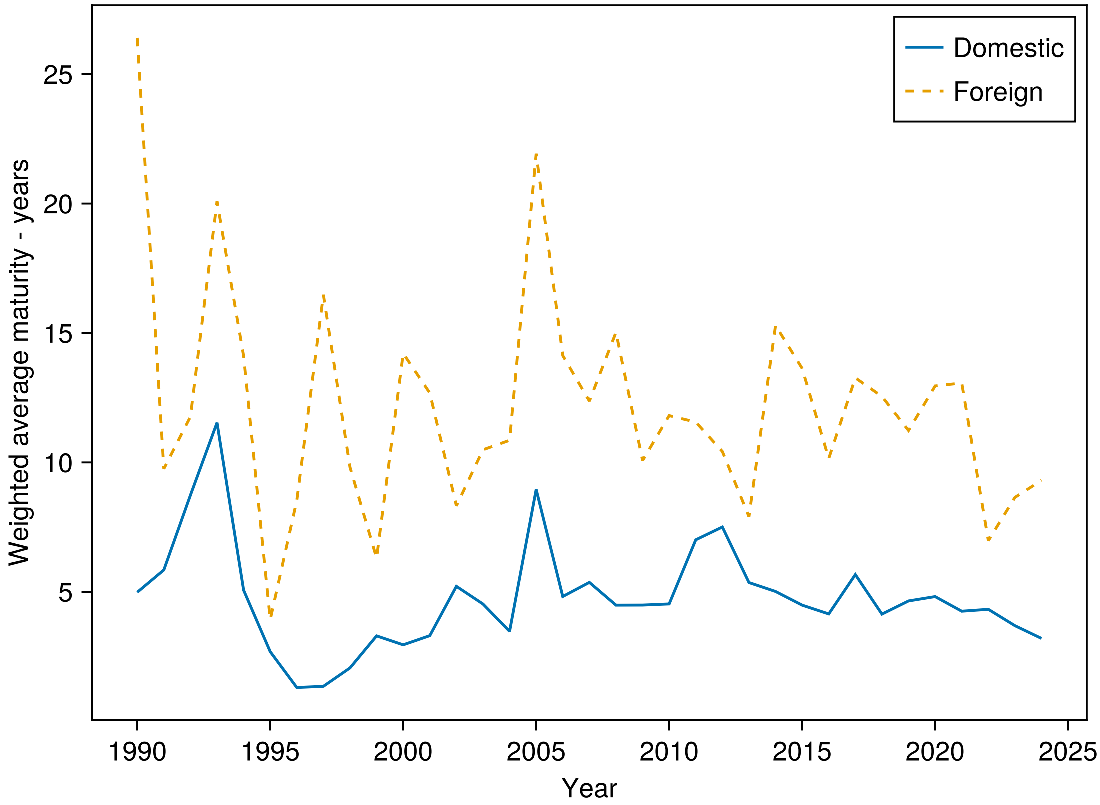


Figure 1: Time Series of Weighted Average Maturity by Currency

tion of Poland, I find that higher spreads are associated with sovereigns tilting their issuance towards more foreign-currency debt. I then pool all the data across countries and run a logit regression of an indicator for domestic-currency issuance on the spread.[6]

The estimate of -0.023 on the spread coefficient is statistically significant at the 1% level, and indicates that a one-percentage-point increase in the spread is associated with a 2.2% decline in the probability of the sovereign issuing a bond denominated in its own currency (equivalently, a roughly 0.2 percentage-point decline in the probability of domestic issuance evaluated at the sample mean), controlling for country and year fixed effects.

Next, I consider the relationship between spreads and the maturity of debt issued. The next three columns of Table 4 show the correlation between spreads and the amount-weighted average maturity of all debt, domestic-currency debt, and foreign-currency debt, respectively. I find that as spreads increase, sovereigns shorten the maturity of their debt issuance. While this effect is present for most countries across domestic currency debt, it is stronger for foreign currency debt.

Finally, Table 5 reports the results of regressing the amount-weighted average maturity of all debt, domestic-currency debt, and foreign-currency debt on the spread, controlling for country and year fixed effects. I find that a one-percentage-point increase in spreads significantly leads to maturities decreasing across all three specifications, although the effect is

[6] If the spread is not available for a given day, I use the nearest available date for which data is available.

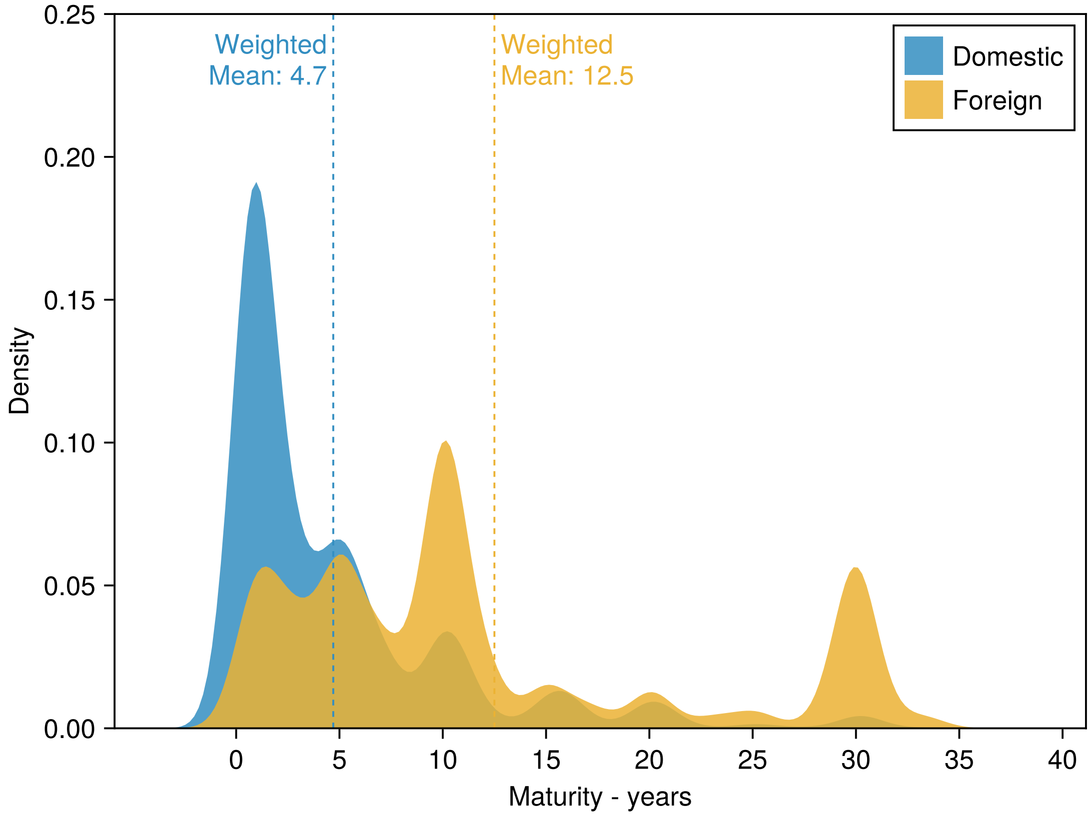


Figure 2: Distribution of Maturity by Currency

almost twice as large in magnitude for foreign currency debt than for domestic currency debt.

II.D Summary

In summary, I use microdata on bonds issued by emerging market economies and spreads to document the following three stylized facts. First, sovereign debt denominated in a country's domestic currency tends to be of much shorter maturity than its debt denominated in a foreign currency. Second, in times of economic distress, sovereigns are more likely to issue debt denominated in a foreign currency. Third, sovereigns also lower the maturity of all their issued debt during times of crises, although this effect is more pronounced for foreign currency debt.

My second and third stylized facts are not completely new to the literature, but they do extend the existing analysis. Ottonello and Perez (2019) show that the domestic currency share of debt is procyclical with respect to the business cycle, while Maeng (2025) finds that the stock of domestic currency debt is negatively related with sovereign's borrowing costs. Arellano and Ramanarayanan (2012) and Broner et al. (2013) document that higher spreads are associated with shorter maturities of debt issue, but they only consider debt denominated in foreign currency.

Meanwhile, the first stylized fact is new to the literature, and its consistency across both time and countries suggests that this is a fundamental feature of sovereign debt markets, and that sovereigns do consider both currency and maturity margins jointly when making their issuance decisions.

Country	Domestic Share	Maturity (All)	Maturity (Domestic)	Maturity (Foreign)
Colombia	-0.090	-0.055	0.016	-0.388
Egypt	-0.386	0.241	0.136	-0.164
Hungary	-0.091	-0.064	-0.085	0.132
Indonesia	-0.135	-0.051	-0.125	-0.117
Jamaica	-0.136	0.047	-0.023	0.065
Mexico	-0.091	-0.124	-0.097	-0.195
Philippines	-0.147	-0.185	-0.179	-0.436
Poland	0.169	-0.350	-0.318	-0.123
Romania	-0.042	-0.323	-0.392	-0.314
Russia	-0.129	-0.274	-0.306	-0.031
Türkiye	-0.224	-0.067	-0.032	-0.277
Uruguay	-0.384	-0.172	-0.055	-0.275

Table 4: Correlation between spreads and currency share and maturity of issuance

	(1)	(2)	(3)	(4)
	$\mathbb{1}$ {Domestic Currency}	Maturity (Years)		
	(Logit)	All	Domestic	Foreign
Spread	-0.023^{***}	-0.054^{**}	-0.050^{**}	-0.097^{*}
	(0.005)	(0.022)	(0.023)	(0.053)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	12,696	1,077	1,018	561
R^2	–	0.271	0.251	0.245

Notes: Column (1) reports a logit regression of an indicator for own-currency issuance on the prevailing spread, estimated at the bond-issuance level. Columns (2)-(4) report OLS regressions of the amount-weighted average maturity of debt issued in a country-quarter on the quarterly-average spread, for all debt, own-currency debt only, and foreign-currency debt only, respectively. All regressions include country and year fixed effects. Standard errors in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 5: Association between spreads and portfolio composition

III Model

In this section, I present a dynamic, small open economy model of defaultable debt that includes bonds with different maturities and currency denomination. There are two types of agents. Firstly, there is a benevolent sovereign borrower who cannot commit to repaying its debts in future periods, nor its future choices over borrowing and monetary policy. Secondly, there is a unit-mass of perfectly competitive and risk-neutral international investors, and they are the sovereign's only source of financing.

III.A Environment

The sovereign is subject to two exogenous shocks: its endowment ($\mathcal{Y}_t$) and the real exchange rate ($\mathcal{E}_t$). I define $\mathcal{E}_t$ as the price of foreign currency in units of domestic currency.[7] Both shocks follow the following process:

$$\begin{pmatrix} \log(\mathcal{Y}_t) \\ \log(\mathcal{E}_t) \end{pmatrix} = \begin{pmatrix} \rho_y & 0 \\ 0 & \rho_e \end{pmatrix} \begin{pmatrix} \log(\mathcal{Y}_{t-1}) \\ \log(\mathcal{E}_{t-1}) \end{pmatrix} + \begin{pmatrix} \varepsilon_t^y \\ \varepsilon_t^e \end{pmatrix} \quad \text{where} \quad \begin{pmatrix} \varepsilon_t^y \\ \varepsilon_t^e \end{pmatrix} \sim N\left(0, \begin{pmatrix} \sigma_y^2 & \rho_{y,e}\sigma_y\sigma_e \\ \rho_{y,e}\sigma_y\sigma_e & \sigma_e^2 \end{pmatrix}\right).$$

ρ_y and ρ_e are the persistence parameters of the endowment and real exchange rate respectively, while σ_y and σ_e denote the volatility of the two shocks. Finally, $\rho_{y,e}$ is the correlation between the two shocks, which is typically negative for emerging market economies.[8]

III.B Sovereign Borrower

As is standard in the literature, I focus on Markov-perfect equilibria and represent the sovereign's infinite-horizon decision problem as a recursive dynamic programming problem.

Every period, the sovereign observes the realized value of its endowment and real exchange rate. It then computes the value of repaying all its debt obligations and defaulting on them, and chooses the option that provides it the most value. For notational convenience, $\mathbb{B}_t = (\{b_t^{SD}, b_t^{SF}, b_t^{LD}, b_t^{LF}\})$ denotes the vector of period t debt obligations for each type of bond, and $\mathbb{X}_t = (\mathcal{E}_t, \mathcal{Y}_t)$ denotes the vector of all exogenous state variables at time t.

The sovereign's value function is given by:

$$V(\mathbb{B}_t, \mathbb{X}_t) = \max_{\{D,R\}} \left\{ V^D(\mathbb{X}_t), V^R(\mathbb{B}_t, \mathbb{X}_t) \right\}. \tag{1}$$

I define the sovereign's policy function for defaulting with the indicator function:

$$d(\mathbb{B}_t, \mathbb{X}_t) = \mathbb{1}\left\{ V^R(\mathbb{B}_t, \mathbb{X}_t) < V^D(\mathbb{X}_t) \right\}. \tag{2}$$

III.B.1 Value of Defaulting

If the sovereign defaults, then all its debt obligations are permanently erased from its resource constraint.[9] However, defaulting comes at two costs. Firstly, the sovereign will suffer output costs given by $h(\mathcal{Y}_t)$. Secondly, the sovereign will be excluded from international financial markets and will not be able to borrow from international investors. It will only regain access with i.i.d. probability θ. Once the sovereign regains access to financial markets, it does so with zero outstanding debt.

Together, the value of defaulting is given by:

$$V^D(\mathbb{X}_t) = u(h(\mathcal{Y}_t)) + \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[\theta V(\mathbf{0}, \mathbb{X}_{t+1}) + (1-\theta) V^D(\mathbb{X}_{t+1}) \right]. \tag{3}$$

[7] $\mathcal{E}_t$ increasing corresponds to the domestic currency depreciating.

[8] I present a stationary version of the model where the price level and nominal value of domestic currency debt are detrended by the previous period's price level.

[9] I do not allow the sovereign to selectively default on only some of its debt obligations.

III.B.2 Value of Repaying

The sovereign's value of repaying its debt obligations is given as follows:

$$V^R(\mathbb{B}_t, \mathbb{X}_t) = \max_{\{\mathbb{B}_{t+1}, \Pi_t\}} \left\{ u(C_t) - l(\Pi_t - \Pi^*) + \beta \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[V(\mathbb{B}_{t+1}, \mathbb{X}_t) \right] \right\} \tag{4}$$

subject to the resource constraint (in domestic currency):

$$\begin{aligned} C_t + \mathcal{E}_t \left[\kappa^S b_t^{SF} + \kappa^L b_t^{LF} \right] + \frac{\kappa^S b_t^{SD} + \kappa^L b_t^{LD}}{\Pi_t} &= q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t) b_{t+1}^{SD} + q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t) \left[b_{t+1}^{LD} - (1-\delta) \frac{b_t^{LD}}{\Pi_t} \right] \\ &+ \mathcal{E}_t \left[q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t) b_{t+1}^{SF} + q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t) \left[b_{t+1}^{LF} - (1-\delta) b_t^{LF} \right] \right] + \mathcal{Y}_t \end{aligned} \tag{5}$$

The sovereign has time separable preferences over consumption, C_t, and gross inflation Π_t, and discounts the future at the constant rate $\beta \in (0, 1)$. As is standard, $u : \mathbb{R}_+ \to \mathbb{R}$ is strictly increasing and concave. To capture the distortional real effects of inflation, I assume that the sovereign suffers disutility from inflation, $l(\Pi_t - \Pi^*)$, where Π^* is the sovereign's target inflation rate and $l : \mathbb{R}_+ \to \mathbb{R}$ is strictly increasing and convex.[10]

To finance consumption and honour its debt obligations, the sovereign can use a combination of (i) inflation to dilute the real cost of its debts denominated in its domestic currency, and (ii) issuing new debt. The sovereign can issue four types of bonds - short-term domestic currency (b_{t+1}^{SD}) and foreign currency (b_{t+1}^{SF}) bonds, and long-term domestic currency (b_{t+1}^{LD}) bonds and foreign currency (b_{t+1}^{LF}) bonds. Both short-term contracts specify prices q^{SD} and q^{SF}, and respective face values b_{t+1}^{SD} and b_{t+1}^{SF}, such that the sovereign receives $q^{SD} b_{t+1}^{SD}$ and $\mathcal{E}_t q^{SF} b_{t+1}^{SF}$ in domestic currency today, and pays $\frac{b_{t+1}^{SD}}{\Pi_{t+1}}$ and $\mathcal{E}_{t+1} b_{t+1}^{SF}$ in domestic currency the next period (conditional on not defaulting).

I model each bond's maturity in the same manner as Chatterjee and Eyigungor (2012), where for each unit of debt, fraction $\delta \in (0, 1]$ matures and a coupon of κ is paid. Short-term debt matures in 1 period and corresponds to $\delta = 1$. The real value of debt denominated in the sovereign's domestic currency can be altered via inflation, whereas the real value of foreign currency bonds is exogenous to the sovereign and increases with $\mathcal{E}_t$.

III.C Foreign Investors

Investors are risk-neutral and perfectly competitive and can access a riskless security that yields a constant gross interest rate of R^*. Hence, investors break even in expectation, yielding the following pricing functions for each type of bond:

$$q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*} \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[d(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \kappa^S \right], \tag{6}$$

$$q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*} \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[d(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \left[\kappa^L + (1-\delta) q^{LF}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1}) \right] \right], \tag{7}$$

$$q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*} \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[d(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \times \frac{1}{\Pi_{t+1}} \times \kappa^S \right], \tag{8}$$

[10] The economy I present in the main text is stationary as I de-trend all domestic currency variables by dividing it by the previous period's price level.

$$q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*}\mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t}\left[d(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \times \frac{1}{\Pi_{t+1}} \times \left[\kappa^L + (1-\delta)q^{LD}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1})\right]\right]. \tag{9}$$

For lenders, all bonds are subject to default risk as given by the sovereign's default policy function $d(\mathbb{B}_{t+1}, \mathbb{X}_{t+1})$. Moreover, bonds denominated in the sovereign's domestic currency entail two additional sources of risk for investors relative to their foreign currency counterparts. Firstly, holding domestic currency bonds exposes investors to fluctuations in the real exchange rate. Secondly, since the sovereign can manipulate the real value of its domestic currency debt via inflation, lenders also need to account for expected inflation in the next period.

Meanwhile, since long-term bonds are repaid to investors over multiple periods, holding them exposes investors to an additional source of risk relative to short-term bonds (holding denominated currency constant), which is dilution risk. Due to the sovereign's inability to commit to future borrowing (and monetary policy), investors are concerned that the sovereign may issue new debt in the future, which would dilute the real value of their existing debt holdings.

All together, the short-term foreign currency bond only needs to compensate investors for default risk, while on the other extreme, the long-term, domestic currency bond needs to compensate investors for risks arising from default, exchange rate fluctuations, inflation, and dilution.

III.D Equilibrium

Definition 1 (Markov Perfect Equilibrium)**.** *A Markov-perfect recursive equilibrium for this economy consists of a set of:*

(i) value functions $\left\{V(\mathbb{B}_t, \mathbb{X}_t), V^D(\mathbb{X}_t), V^R(\mathbb{B}_t, \mathbb{X}_t)\right\}$,

(ii) pricing functions $\left\{q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t)\right\}$, *and*

(iii) policy functions $\left\{\hat{b}^{LD}(\mathbb{B}_t, \mathbb{X}_t), \hat{b}^{LF}(\mathbb{B}_t, \mathbb{X}_t), \hat{b}^{SD}(\mathbb{B}_t, \mathbb{X}_t), \hat{b}^{SF}(\mathbb{B}_t, \mathbb{X}_t), \hat{\Pi}_{t+1}(\mathbb{B}_t, \mathbb{X}_t)\right\}$

such that the following conditions hold:

1. *Taking the bond price functions as given, the sovereign's policy functions and value functions solve the sovereign's dynamic programming problem, and*

2. *The bond price functions* $q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t)$ *satisfy* (6), (7), (8), *and* (9) *respectively.*

IV Optimality Conditions

In this section, I discuss the economic trade-offs associated with the sovereign's optimal debt portfolio. Ultimately, I argue that the short-term, foreign currency bond offers the most disciplining benefits out of all four bonds, but at the cost of having no hedging benefits. Meanwhile, a bond being longer-term or being denominated in domestic currency offers hedging benefits, but at the cost of having weaker disciplining benefits. This culminates in the long-term domestic currency bond having the most hedging benefits, but at the cost of possessing the heaviest incentive costs.

This section is organized as follows. I first present the inter-temporal optimality condition of each bond corresponding to the full problem. Then, in order to show the trade-offs between each bond more transparently, I use the short-term, foreign currency bond as a baseline since it is the simplest and most understood, and I vary one margin at a time. The first margin I vary is currency denomination, which I discuss in subsection IV.A. Then, I only consider foreign currency bonds and vary the maturity of the bond in subsection IV.B. Finally, I vary both margins simultaneously by comparing the short-term, foreign currency bond against the long-term, domestic currency bond in subsection IV.C. To ease the notational burden, I set $\kappa^S = \kappa^L = 1$ without loss of generality.

Let $R(\mathbb{B}_t)$ denote the set of $\{\mathcal{E}_t, \mathcal{Y}_t\}$ realizations for which the sovereign chooses to repay its debts:

$$R(\mathbb{B}_t) = \left\{\mathcal{E}_t, \mathcal{Y}_t : V^R(\mathbb{B}_t, \mathbb{X}_t) \geq V^D(\mathbb{X}_t)\right\} \tag{10}$$

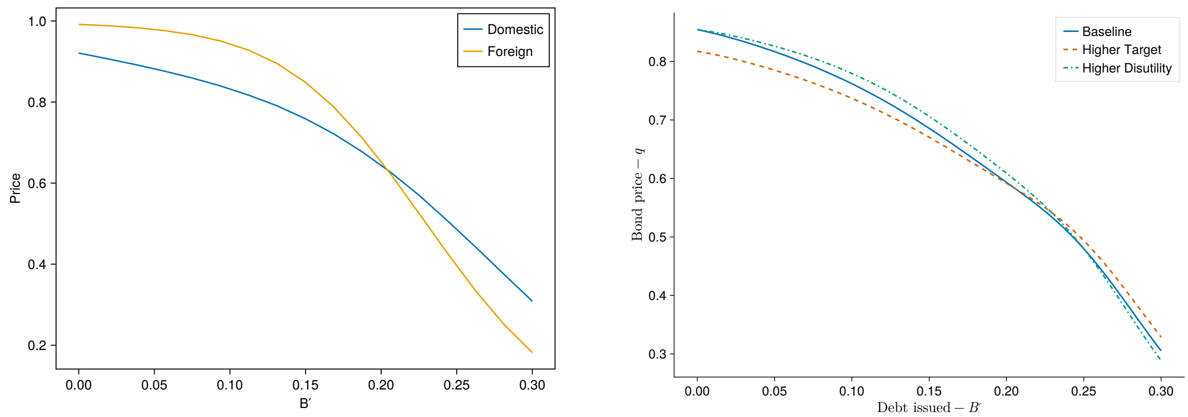
Assuming repayment, the first order necessary conditions for the sovereign's optimal portfolio choice are given by the following four equations, which correspond to the first-order conditions of (4) with respect to each bond's issuance decision.

$$u_C(C_t)\left[q^{SF} + \frac{\partial q^{SF}}{\partial b^{SF}_{t+1}} b^{SF}_{t+1} + \frac{\partial q^{LF}}{\partial b^{SF}_{t+1}}\left[b^{LF}_{t+1} - (1-\delta)b^{LF}_{t+1}\right] + \frac{\partial q^{SD}}{\partial b^{SF}_{t+1}} b^{SD}_{t+1} + \frac{\partial q^{LD}}{\partial b^{SF}_{t+1}}\left[b^{LD}_{t+1} - (1-\delta)\frac{b^{LD}_t}{\Pi_t}\right]\right] \tag{11}$$
$$= \beta \int_{R_{t+1}} u_C(C_{t+1})\mathcal{E}_{t+1} f(\mathbb{X}_t, \mathbb{X}_{t+1}) d\mathbb{X}_{t+1}$$

$$u_C(C_t)\left[q^{SD} + \frac{\partial q^{SF}}{\partial b^{SD}_{t+1}} b^{SF}_{t+1} + \frac{\partial q^{LF}}{\partial b^{SD}_{t+1}}\left[b^{LF}_{t+1} - (1-\delta)b^{LF}_{t}\right] + \frac{\partial q^{SD}}{\partial b^{SD}_{t+1}} b^{SD}_{t+1} + \frac{\partial q^{LD}}{\partial b^{SD}_{t+1}}\left[b^{LD}_{t+1} - (1-\delta)\frac{b^{LD}_t}{\Pi_t}\right]\right] \tag{12}$$
$$= \beta \int_{R_{t+1}} u_C(C_{t+1}) \frac{1}{\Pi_{t+1}} f(\mathbb{X}_t, \mathbb{X}_{t+1}) d\mathbb{X}_{t+1}$$

$$u_C(C_t)\left[q^{LF} + \frac{\partial q^{SF}}{\partial b^{LF}_{t+1}} b^{SF}_{t+1} + \frac{\partial q^{LF}}{\partial b^{LF}_{t+1}}\left[b^{LF}_{t+1} - (1-\delta)b^{LF}_{t+1}\right] + \frac{\partial q^{SD}}{\partial b^{LF}_{t+1}} b^{SD}_{t+1} + \frac{\partial q^{LD}}{\partial b^{LF}_{t+1}}\left[b^{LD}_{t+1} - (1-\delta)\frac{b^{LD}_t}{\Pi_t}\right]\right] \tag{13}$$
$$= \beta \int_{R_{t+1}} u_C(C_{t+1})\left[1 + (1-\delta)q^{LF}_{t+1}\right]\mathcal{E}_{t+1} f(\mathbb{X}_t, \mathbb{X}_{t+1}) d\mathbb{X}_{t+1}$$

$$u_C(C_t)\left[q^{LD} + \frac{\partial q^{SF}}{\partial b^{SD}_{t+1}} b^{SF}_{t+1} + \frac{\partial q^{LF}}{\partial b^{SD}_{t+1}}\left[b^{LF}_{t+1} - (1-\delta)b^{LF}_{t}\right] + \frac{\partial q^{SD}}{\partial b^{SD}_{t+1}} b^{SD}_{t+1} + \frac{\partial q^{LD}}{\partial b^{SD}_{t+1}}\left[b^{LD}_{t+1} - (1-\delta)\frac{b^{LD}_t}{\Pi_t}\right]\right] \tag{14}$$
$$= \beta \int_{R_{t+1}} u_C(C_{t+1}) \times \frac{1}{\Pi_{t+1}} \times \left[1 + (1-\delta)q^{LD}_{t+1}\right] f(\mathbb{X}_t, \mathbb{X}_{t+1}) d\mathbb{X}_{t+1}$$



(a) Pricing Functions: Currency Choice

(b) Comparative Statics: Currency Choice

Figure 3: Pricing Functions of Domestic and Foreign Currency Debt

The left-hand side of each equation corresponds to the marginal benefit of issuing an additional unit of that bond, while the right-hand side corresponds to the marginal cost of issuing an additional unit of that bond. The marginal benefit is given by the marginal utility of consumption multiplied by the bond's price and its sensitivity to changes in issuance. The terms in the square bracket capture the marginal revenue from issuing an additional unit of that bond, which includes the direct effect of the bond's price and the indirect effect of the bond's price on all other bond prices. Since the incentive to default and inflate next period is increasing with all debt issuance, all bond prices are decreasing with respect to issuance. Meanwhile, the marginal cost is given by the discounted expected marginal utility of consumption in the next period, adjusted for inflation and any future bond prices.

IV.A Margin 1: Currency

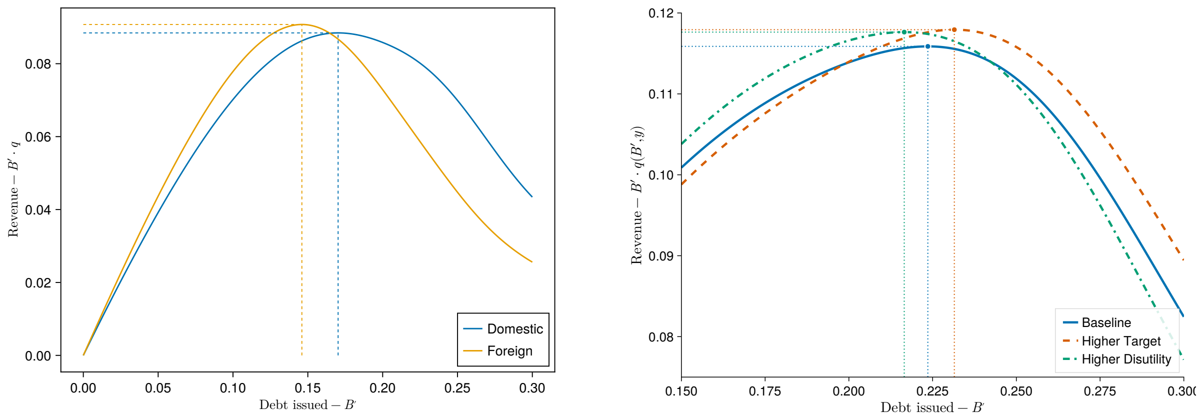
In this subsection, I highlight the economic trade-offs between domestic and foreign currency debt.

IV.A.1 Domestic Currency's Incentive Costs

I first argue that foreign currency debt has disciplining benefits for the sovereign relative to domestic currency debt (holding maturity constant). This can be seen in Figure 3a, where I show the equilibrium pricing functions for domestic and foreign currency debt as a function of the corresponding amount of debt issued while holding the other bond constant at zero. There are two noteworthy aspects.

Firstly, as long as the sovereign's target inflation rate is strictly positive, then investors will anticipate that holding any amount of domestic currency debt will result in its real value being eroded by at least the target inflation rate next period. Meanwhile, the foreign currency bond's real value is not eroded by domestic inflation.

Secondly, the foreign currency bond pricing function is steeper than the domestic currency bond. As reflected in the decreasing slope of the pricing functions, the sovereign's incentive to



(a) Revenue Curves: Currency Choice

(b) Comparative Statics: Currency Choice

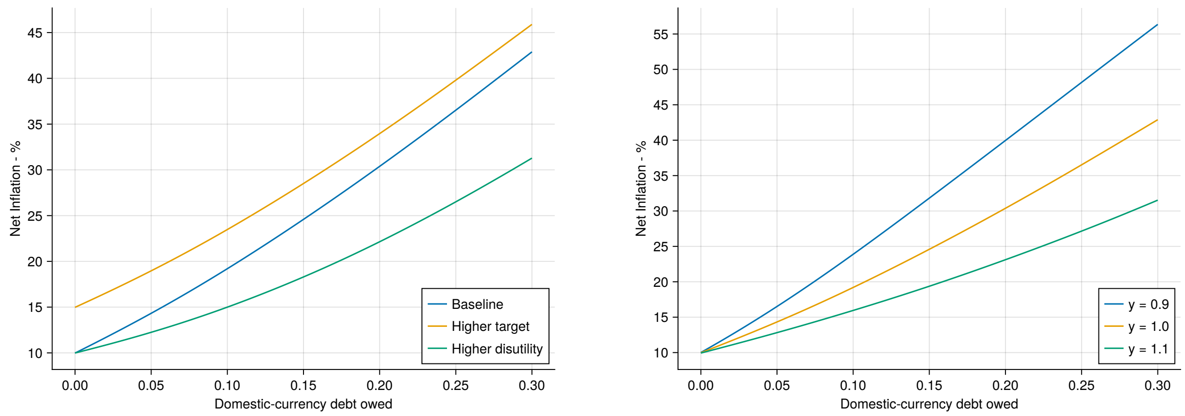
Figure 4: Revenue Curves of Domestic and Foreign Currency Debt

default rises with the amount of debt issued, irrespective of currency denomination. However, default risk rises more quickly with respect to foreign currency than domestic currency debt. This is because as long as the sovereign targets a strictly positive net inflation rate, then it can, at a cost (from a utility perspective), reduce its domestic currency burden, which translates to the sovereign being able to owe a higher amount of domestic currency debt for which it would find it optimal to repay in equilibrium. Thus, while issuing more domestic currency debt increases the sovereign's incentive to default and engage in costly inflation, these incentive costs grow slower with domestic currency debt issuance than foreign currency debt issuance, and the initial premium the sovereign pays from the target inflation rate overtakes the foreign currency bond's higher default risk at $B' = 0.133$.

Figure 3b shows the equilibrium domestic currency pricing function for different values of the sovereign's target inflation rate and inflation disutility parameter. When the target rate increases, the inital wedge grows, while a high inflation disutility parameter renders the pricing function initially less steep. This is because the more disutility the sovereign incurs from inflation, the less domestic currency debt it can hold before it would find it optimal to default.

The relative incentive benefits of foreign currency debt over domestic currency debt can further be seen in the revenue curves in Figure 4a. I vary the amount of domestic and foreign currency debt issued along the x-axis (while holding the other bond constant at zero), and plot the corresponding revenue generated from issuing that debt along the y-axis. Revenue initially increases with debt issuance. However, due to the decreasing slope of the pricing function, the revenue eventually peaks and then decreases with further debt issuance since the negative price effects dominate. Not only can the sovereign raise more revenue from issuing foreign currency debt than domestic currency debt, but it is also able to accomplish that higher level of revenue with a lower amount of debt issued and thus owed next period.

Note that the revenue functions and pricing functions simply reflect the sovereign's choice set for a given state of the world. In other words, it reflects what options the sovereign can choose among, but which portfolio it ultimately chooses from that menu in equilibrium is



(a) Inflation policy function

(b) Inflation policy function comparative statics

Figure 5: Policy function for inflation

further determined by the sovereign's debt burden.

Lastly, the role of the sovereign's target inflation rate and inflation disutility parameter on revenue generation are shown in Figure 4b. When the target inflation rate increases, the sovereign needs to issue more domestic currency debt to attain its maximum, while increasing the inflation disutility parameter lowers the required amount of debt issuance. Meanwhile, how varying the two parameters affects the maximum revenue further depends on the sovereign's value function for defaulting, and especially the output costs from defaulting.

In this example, the sovereign's output costs from defaulting are relatively low, so default risk increases faster with debt issued, resulting in the flatter pricing function from the higher inflation target dominating the initial wedge. However, if the sovereign's output costs from defaulting were higher, the default risk would grow more slowly, and a higher target inflation rate may result in a lower maximized revenue.

IV.A.2 Domestic Currency's Hedging Benefits

I now show that on the other hand, domestic currency debt provides the sovereign with hedging benefits. The marginal cost of foreign and domestic currency debt is given by the right hand side of (11) and (12). This is where the real exchange rate being negatively correlated with real endowment becomes crucial. In such a world, the sovereign's marginal utility of consumption would covary positively with the real cost of repaying foreign currency debt. In other words, repaying foreign currency debt is more costly when the sovereign is in a bad state with low endowment.

Meanwhile, domestic currency debt would have insurance properties if the inverse of next period's inflation co-moved negatively with the marginal utility of consumption. In a currency-choice problem where all debt is short-term, the sovereign's static optimality condition for inflation is:

$$u_C(C_t)\left[\frac{1}{\Pi_t^2}\right]\left[b_t^{SD}\right] = l_\Pi(\Pi_t - \Pi^*). \tag{15}$$

I present the equilibrium policy function for inflation as the amount of domestic currency debt owed increases in Figure 5a. As long as any strictly positive amount of domestic currency debt is owed, then the sovereign is able to use inflation to relax its real resource constraint via diluting the real cost of repaying its domestic currency debt, thereby enjoying higher consumption. Optimal inflation equates the marginal utility gain from higher consumption with the marginal disutility from deviating from the target inflation rate.

In order to understand the economic role of the sovereign's target inflation rate, Π^*, and disutility parameter, ψ, I plot the different inflation policy functions across different parameter values (holding the state constant) in Figure 5a. When the sovereign's target inflation rate is higher, then the sovereign is able set a higher level of inflation without incurring any disutility. However, note that the policy function does not shift linearly with the change in the target inflation rate as for $b_t^{SD} > 0$. This is because as the target inflation rate increases, the marginal utility gain from higher consumption increases more slowly with domestic currency debt owed, resulting in a steeper policy function.

If the sovereign's disutility parameter is higher, optimal inflation at $b_t^{SD} = 0$ is unchanged since the sovereign is not able to use inflation to relax its real resource constraint. However, as the amount of domestic currency debt owed increases, the marginal disutility from inflation increases faster, resulting in a flatter policy function with respect to domestic currency debt obligations.

Finally, Figure 5b shows that holding the amount of domestic debt constant, optimal inflation is decreasing with respect to endowment. If the sovereign is suffering a recession where GDP is low, then its marginal utility of consumption is higher, and the marginal benefit of relaxing the resource constraint via inflation is larger, resulting in higher inflation.

IV.B Margin 2: Maturity

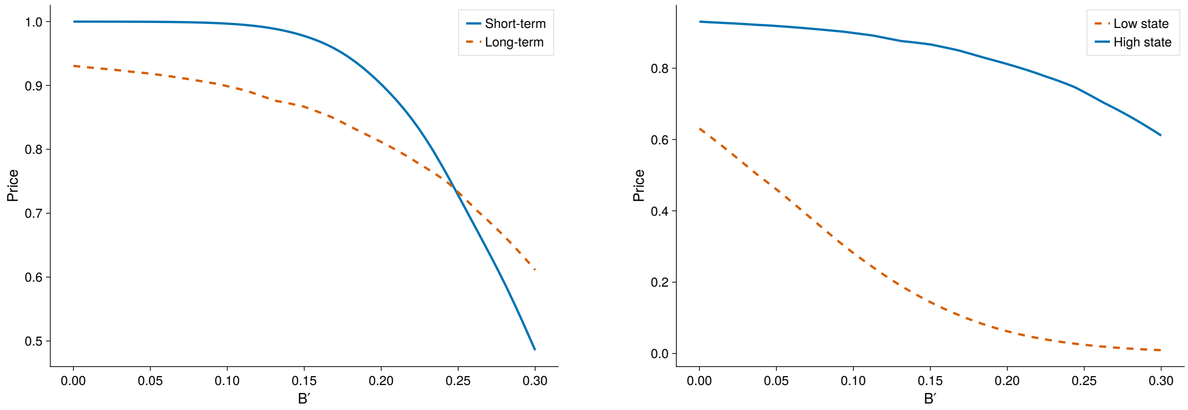
Now, I highlight the hedging-incentive trade-offs that arise when the sovereign chooses between short-term and long-term debt. Since the margin of adjustment is now maturity, the analysis in this subsection assumes that all debt is denominated in foreign currency, and that the real exchange rate is constant at unity.[11].

IV.B.1 Long Term Debt's Incentive Costs

Issuing long-term debt is costly from a disciplining perspective because the sovereign lacks commitment in its future actions. While the sovereign may choose a b_{t+1}^{LF} such that default risk next period is low, it cannot promise that it will not choose a b_{t+2}^{LF} that increases default risk in the subsequent period. This dilutes the market value of the unmatured long-term debt.

Forward-looking investors anticipate this dilutionary incentive and factor it into their pricing, resulting in the pricing functions in Figure 6a. This translates to the revenue curves in Figure 7. Figure 7a displays the revenue curves for the short-term and long-term debt separately, showing that short-term debt is able to generate higher revenue at lower levels of debt issuance compared to long-term debt. I further plot the equilibrium revenue curves as I vary

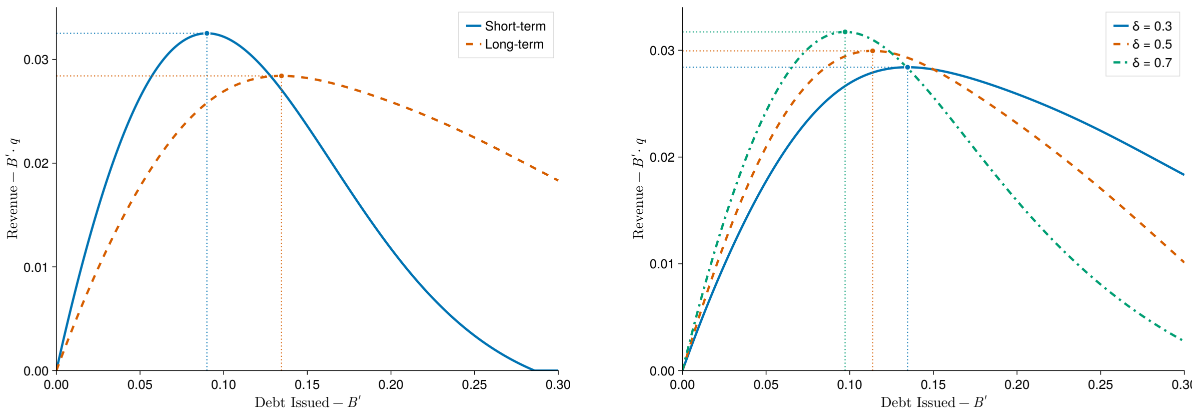
[11] The parameter values for this exercise mirror those used in Arellano and Ramanarayanan (2012). I set $\lambda_0 = -0.51$ and $\lambda_1 = 0.53$, which approximately resemble the threshold-cost specification in Arellano and Ramanarayanan (2012)



(a) Pricing Functions: Maturity Choice

(b) Comparative Statics: Maturity Choice

Figure 6: Pricing Functions of Short and Long Term Debt



(a) Revenue Curves: Maturity Choice

(b) Comparative Statics: Maturity Choice

Figure 7: Revenue Curves of Short and Long Term Debt

the maturity of the long-term bond in Figure 7b. As δ decreases (or the maturity of the long-term bond increases), the relative incentive disadvantage of long-term debt over short-term debt becomes more pronounced.

IV.B.2 Long Term Debt's Hedging Benefits

While long-term debt is costlier relative to short-term debt, it provides the sovereign hedging benefits. This can be seen by comparing the right hand side of (11) and (13) - the real cost of repaying the short-term bond is invariant to the sovereign's state of the world, while the real cost of repaying the long-term bond would covary negatively with the sovereign's marginal utility of consumption if q^{LF} decreased with $\mathcal{Y}_t$.

This is indeed the case, as shown in Figure 6b, which shows q^{LF} for a low-endowment and high-endowment state. In bad states of the world, the market value of the unmatured long-term debt is lower as future default risk is heightened. This means that conditional on repayment, the cost of serving the long-term debt is lower in bad states of the world.

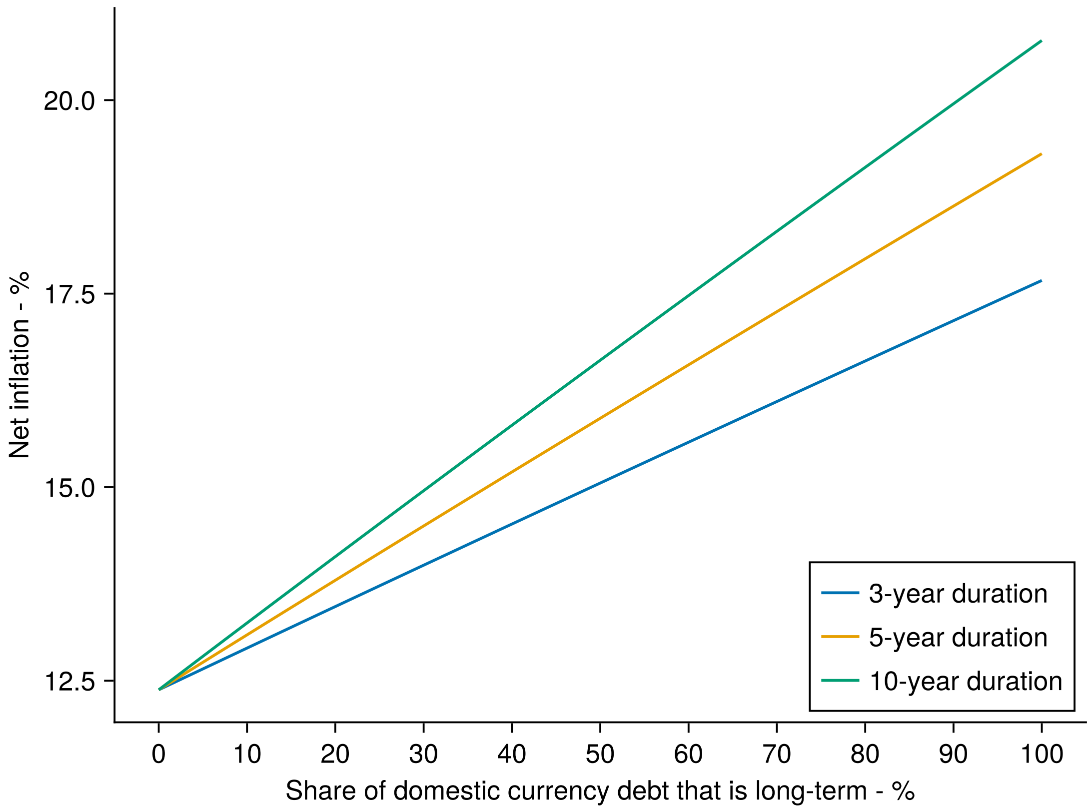


Figure 8: Maturity's effects on optimal inflation

IV.C Both Margins

Lastly, I now compare the short-term, foreign currency bond against the long-term, domestic currency bond. In the full problem, the sovereign's intra-optimality condition for inflation is:

$$u_C(C_t)\left[\frac{1}{\Pi_t^2}\right]\left[b_t^{LD} + b_t^{SD} + q^{LD}(1-\delta)b_t^{LD}\right] = l_\Pi(\Pi_t - \Pi^*). \tag{16}$$

There are two important differences between (16) and (15), which is the optimality condition for inflation when all debt is short-term. First, holding the amount of domestic currency debt constant, optimal inflation is increasing in the share of that debt that is long-term. In other words, one unit of long-term domestic debt increases optimal inflation more than one unit of short-term domestic debt. Second, holding all else constant, the optimal inflation is increasing with respect to the maturity of the long-term domestic currency bond.

These two forces only arise when both currency and maturity are examined simultaneously. When the domestic currency debt is long term, the sovereign can use inflation to reduce the real value of its debt over time. Inflation not only erodes the real value of the fraction that is due today, but also the unmatured fraction that is due in the future. The longer-term the domestic currency bond (the lower δ is), the greater the sovereign is able to use inflation to reduce its real value over time.

These two forces are shown in Figure 8, where I plot the optimal inflation rate while varying the share of domestic currency debt that is long-term (while holding the total amount of domestic currency debt constant). I then also vary the maturity of the long-term domestic currency bond relative to the short-term domestic currency bond. The longer-term the bond is, the greater the sovereign is incentivized to inflate to erode future debt obligations as well.

Thus, issuing a unit of the long-term domestic currency bond increases investors' expectations of future inflation more than an equivalent unit of short-term domestic currency debt. When coupled with the aforementioned dilution risk present in all long-term debt as shown in subsection IV.B, they culminate in the long-term domestic currency bond having the worst incentive costs of all four bonds.

On the other hand, the individual insurance benefits of domestic currency and long-term debt also compound to give the long-term domestic currency bond the most hedging benefits of all four bonds. As the right hand side of (14) shows, in bad states of the world where the marginal utility of consumption is high, the real cost of repaying the long-term domestic currency bond decreases due to both inflation and the lower market value of the unmatured long-term domestic currency bond.

V Calibration

Parameter	**Description**	**Value**	**Source**
σ	risk aversion	5	standard value
R^*	risk-free rate	1.04	1-year US Treasury yield
δ	perpetuity decay factor	0.1	10 year maturity
κ^S	ST coupon payments	1.04	Normalization
κ^L	LT coupon payments	0.14	Normalization
θ	re-entry probability	0.17	6-year exclusion
ρ_y	output persistence	0.70	data
σ_y	output volatility	0.028	data
ρ_e	real exchange rate persistence	0.85	data
σ_e	real exchange rate volatility	0.09	data
$\rho_{y,e}$	shock correlation	-0.32	data
α	taste shock variance	0.01	numerical convergence

Table 6: Parameters Selected Directly

I solve the model numerically via value function iteration to evaluate its quantitative predictions on the sovereign's dynamic portfolio choice of debt issuance.

As shown in Chatterjee and Eyigungor (2012), converge is notoriously difficult to achieve in sovereign debt models with long-term debt without any randomization. I thus follow Mateos-Planas et al. (2026) and assume that the sovereign's default-repayment decision is subject to additive taste shocks that are i.i.d. and Type-1 Extreme-Value (Gumbel) distributed. Specifically, I set the location parameter to $-\alpha(\gamma+\ln(2))$ where γ is the Euler-Mascheroni constant, and α is the variance of the shocks. The choice of the location parameter ensures that

Parameter	Description	Value
β	discount factor	0.92
λ_0	default cost - linear	-0.71
λ_1	default cost - quadratic	0.77
ψ	inflation dis-utility	1.1
Π^*	inflation target	9.5

Table 7: Parameters Selected By Matching Moments

Statistics	Data	Model
Mean debt:GDP ratio	12.6%	8.9%
Mean spread (1 year, domestic currency)	10.26%	12.8%
Std. spread (1 year, domestic currency)	2.97%	4.1%
Mean inflation	8.59%	11.52%
Std. inflation	1.82%	1.6%

Table 8: Model Results and Data Moments

Statistics	Data	Model
corr(NX/y, y)	-0.12	-0.26
Domestic currency share	89.66%	77.4%
Mean maturity (domestic currency)	3.12 years	2.2 years
Mean maturity (foreign currency)	8.61 years	5.9 years
Mean spread (1 year, foreign currency)	2.55%	2.8%
Mean spread (10 year, foreign currency)	4.74%	6.3%
Mean spread (10 year, domestic currency)	7.16%	17.1%

Table 9: Untargeted Moments

the taste shocks have mean zero and thus does not give any ex-ante utility to either default or repayment.

Hence, the sovereign's value function ex-ante of taste shocks is given by:

$$\begin{aligned} V^O(\mathbb{B}_t, \mathbb{X}_t) &= \mathbb{E}_{\varepsilon_t^D, \varepsilon_t^R}\left[V^D(\mathbb{X}_t) + \varepsilon_t^D, V^R(\mathbb{B}_t, \mathbb{X}_t + \varepsilon_t^R)\right] \\ &= \alpha \log\left[\exp\left(\frac{V^D(\mathbb{X}_t)}{\alpha}\right) + \exp\left(\frac{V^R(\mathbb{B}_t, \mathbb{X}_t)}{\alpha}\right)\right] - \alpha \ln(2). \end{aligned} \tag{17}$$

This assumption replaces the sovereign's deterministic default-repayment decision with a probabilistic one, which smooths the bond pricing functions and yields a closed-form expression for the ex-ante probability of repayment (before observing the taste-shock realization):

$$P(d_t = 0 | \mathbb{B}_t, \mathbb{X}_t) = \frac{\exp\left(\frac{V^R(\mathbb{B}_t, \mathbb{X}_t)}{\alpha}\right)}{\exp\left(\frac{V^D(\mathbb{X}_t)}{\alpha}\right) + \exp\left(\frac{V^R(\mathbb{B}_t, \mathbb{X}_t)}{\alpha}\right)}. \tag{18}$$

Two further numerical challenges arise when solving the model. Firstly, due to the curse of dimensionality, conventional discrete-choice grid based methods are computationally infeasible.[12] Secondly, the state-space over which the policy functions for bond issuance are well-defined is irregular. To overcome these challenges, I use Gaussian Process Regression to approximate all equilibrium functions in the model. I provide details of my numerical algorithm in Appendix A.

I discipline my model by calibrating it to the Turkish economy over the period 2004:I-2018:II. My sample period begins in 2004:I since the Turkish lira only became free floating after 2001, and Turkish inflation only became relatively stable from 2004 onwards. I end in 2018:II since that was when the Great Turkish depression began.

One period in my model corresponds to one year. I impose the following functional form for sovereign's period utility function, which is separable in consumption and inflation:

$$u(C_t) - l(\Pi_t - \Pi^*) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\psi}{2}(\Pi_t - \Pi^*)^2, \tag{19}$$

where σ is the sovereign's coefficient of relative risk aversion, ψ is the dis-utility of inflation, and Π^* is the inflation target. Following Chatterjee and Eyigungor (2012), I assume that the output costs of default are given by the following functional form:

$$h(\mathcal{Y}_t) = \mathcal{Y}_t - \max\left\{0, \lambda_0 \mathcal{Y}_t + \lambda_1^2 \mathcal{Y}_t\right\}. \tag{20}$$

With these functional forms, 17 parameter values must be specified. I summarize parameter values that were selected directly in Table 6 and parameters that were selected by matching moments from the Turkish economy in Table 7.

The parameters of the joint endowment-real exchange process were estimated on linearly detrended, quarterly real GDP and real exchange rate data from FRED, which was then annualized.[13]

I set the risk aversion parameter to $\sigma = 5$, which is among the higher end of values typically used in real business cycles models. If the sovereign is not sufficiently risk averse, then numerical issues of portfolio indifference may arise. I set the variance of the Gumbel-distributed taste shock to $\alpha = 0.01$, which is small enough to ensure that the sovereign's default-repayment decision is not dominated by taste shocks, but large enough for numerical convergence.

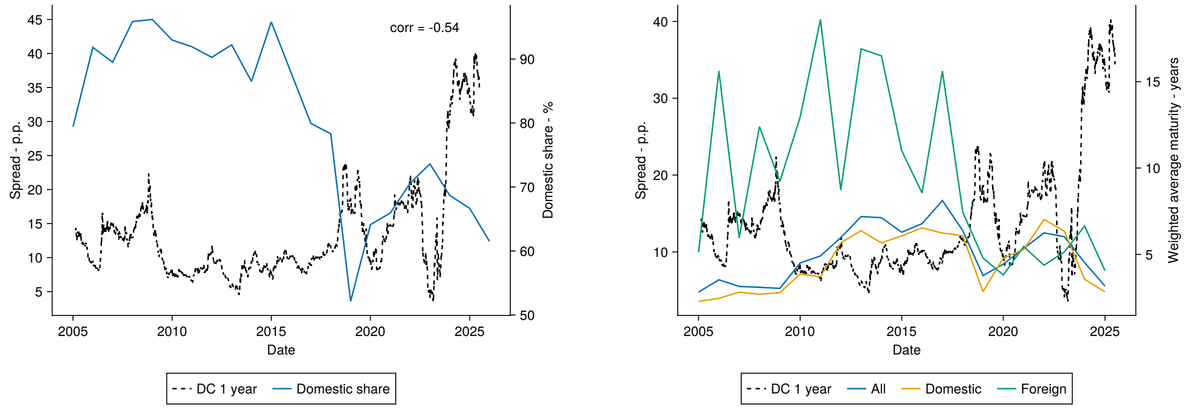
The parameters describing the maturity of the bonds were set as follows. Over 2004:I-2018:II, the average maturity of the foreign currency debt that Türkiye issued was 8.6 years.[14] Since the average maturity of domestic and foreign currency debt that my model generates will be a convex combination of the 1-year and long term bond, I set the decay parameter of the long-term bond to correspond to 10 years in maturity, or $\delta = 0.1$.[15] I do not allow the maturity of the long-term domestic currency bond to be different from the long-term foreign currency

[12]This restriction precludes the use of taste shocks to the sovereign's bond issuance decision, which is why all my policy functions for bond issuance and inflation are deterministic.

[13]I use Cholesky decomposition to transform the correlated bivariate normal variables into two independent normal variables. I then use Gauss-Hermite quadrature with 81 nodes (9 nodes per shock) as a tensor product to compute expected values.

[14]The average maturity of Turkish lira denominated debt over this period was 3.6 years.

[15]Data on domestic currency spreads is also only available for bonds of 1, 3, 5, 7, and 10 years in maturity.



(a) Currency share (b) Weighted average maturity

Figure 9: Time series of Turkiye's issuance portfolio and 1-year spread

bond, since there is no way to obtain a value of the decay parameter for the domestic currency bond from the data in a way that does not bias the model's predictions toward generating a lower average maturity for domestic currency debt.

Following Mihalache (2020), I set the coupon of short-term and long-term debt to $\kappa^S = R^*$ and $\kappa^L = \delta + R^* - 1$, respectively. This has the convenient property of normalizing the risk-free price of both foreign currency bonds to 1, which facilitates easier comparison of bond prices across different maturities.[16]

I follow Arellano and Ramanarayanan (2012) and set the re-entry probability to $\theta = 0.17$, which corresponds to an average exclusion period of 6 years. Finally, the gross risk-free rate was set to $R^* = 1.04$, which is roughly the annualized yield of a 1 year US treasury bond.

The remaining five parameters, β, λ_0, λ_1, ψ, and Π^*, are selected by matching (i) an average external debt to GDP ratio of 13.8%, (ii) an average spread of 10.5% for 1-year bonds issued by the Turkish government, (iii) a standard deviation of 3.3% for spreads, (iv) an average annualized inflation rate of 11.86%, and (iv) a standard deviation of 9.34% inflation. For data on spreads, I use data from Du and Schreger (2016), Du and Schreger (2022) and Du et al. (2025), which provides daily frequency data on the difference between the yield of a Turkish lira denominated bond and a US treasury bond of the same tenor. Data on inflation and the external trade balance were computed from FRED and the World Bank's World Development Indicators, respectively.

VI Quantitative Analysis

VII Conclusion

[16]The risk-free price of a foreign currency bond with decay parameter δ is given by $\frac{\kappa}{\delta + R^* - 1}$.

Statistic	Overall
$\text{corr}(\mathcal{E}_t, u_C(C_t))$	0.32
$\text{corr}(\frac{1}{\Pi_t}, u_C(C_t))$	-0.27
$\text{corr}(1-(1-\delta)q_t^{LF}, u_C(C_t))$	-0.64
$\text{corr}(\frac{1}{\Pi_t} \times (1-(1-\delta)q_t^{LD}), u_C(C_t))$	-0.77

Table 10: Hedging Benefits

References

Arellano, Cristina and Ananth Ramanarayanan, "Default and the Maturity Structure in Sovereign Bonds," *Journal of Political Economy*, April 2012, *120* (2), 187–232.

Arslanalp, Serkan and Takahiro Tsuda, "Tracking Global Demand for Advanced Economy Sovereign Debt," *IMF Working Papers*, 2012, *12* (284), 1.

Broner, Fernando A., Guido Lorenzoni, and Sergio L. Schmukler, "Why Do Emerging Economies Borrow Short Term?," *Journal of the European Economic Association*, January 2013, *11* (suppl_1), 67–100.

Chatterjee, Satyajit and Burcu Eyigungor, "Maturity, Indebtedness, and Default Risk," *American Economic Review*, May 2012, *102* (6), 2674–2699.

Du, Wenxin and Jesse Schreger, "Local Currency Sovereign Risk," *The Journal of Finance*, June 2016, *71* (3), 1027–1070.

_ **and** _ , "Sovereign Risk, Currency Risk, and Corporate Balance Sheets," *The Review of Financial Studies*, September 2022, *35* (10), 4587–4629.

_ **, Ritt Keerati, and Jesse Schreger**, "Decoupling Dollar and Treasury Privilege," *International Finance Discussion Paper*, December 2025, (1427), 1.

Gu, Grace Weishi and Zachary R. Stangebye, "Costly Information and Sovereign Risk," *International Economic Review*, November 2023, *64* (4), 1397–1429.

Gürkaynak, Refet S., Brian Sack, and Jonathan H. Wright, "The U.S. Treasury yield curve: 1961 to the present," *Journal of Monetary Economics*, November 2007, *54* (8), 2291–2304.

Maeng, Fred Seunghyun, "Default, Inflation Expectations, and the Currency Denomination of Sovereign Bonds," Technical Report, Nanyang Technological University November 2025.

Mateos-Planas, Xavier, Sean McCrary, Jose-Vıctor Rıos-Rull, and Adrien Wicht, "The Generalized Euler Equation and the Bankruptcy-Sovereign Default Problem," Technical Report June 2026.

Mihalache, Gabriel, "Sovereign default resolution through maturity extension," *Journal of International Economics*, July 2020, *125*, 103326.

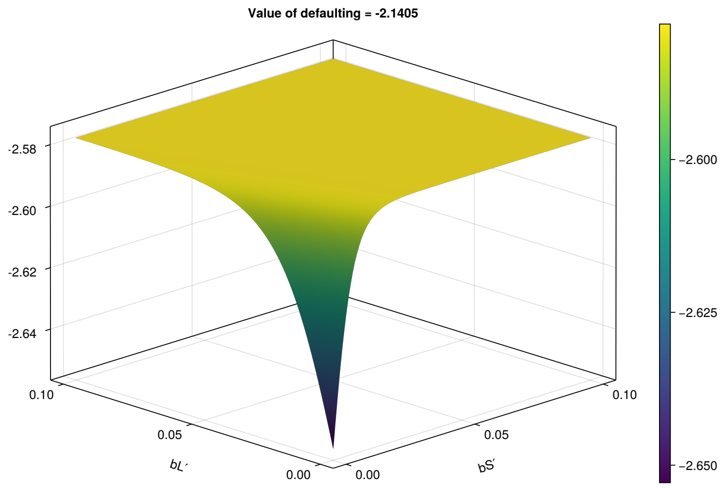


Figure 10: Objective Function Across Different Portfolios in a Default State

Ottonello, Pablo and Diego J. Perez, "The Currency Composition of Sovereign Debt," *American Economic Journal: Macroeconomics*, July 2019, *11* (3), 174–208.

Schwert, Michael, "Municipal Bond Liquidity and Default Risk," *The Journal of Finance*, August 2017, *72* (4), 1683–1722.

A Solution Method

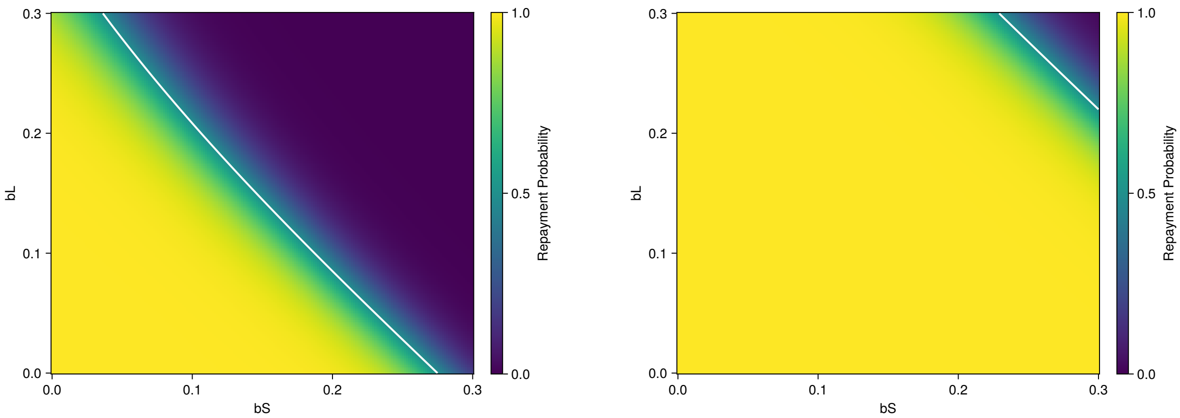
I solve the model with value function iteration, where I approximate all equilibrium functions as Gaussian processes. I use Gaussian process regression for three main reasons. Firstly, the number of points required to accurate approximate the equilibrium functions is orders of magnitude lower than even powerful sparse grid methods such as Smolyak's method.

Secondly, Gaussian process regression inherently incorporates uncertainty quantification, which is particularly useful because Bayesian Active Learning can be used to endogenously add training points in regions of the state space where the Gaussian process is most uncertain. In highly non-linear models such as this one, this adaptivity is extremely useful.

Lastly, and most importantly, Gaussian process regression is grid-less. This is particularly crucial when approximating the policy functions for bond issuance becuase the domain over which the policy functions are well-defined is irregular.

Figure 10 shows the objective function in the maturity choice model of Arellano and Ramanarayanan (2012) evaluated at across a fine grid of potential short-term and long-term bond portfolio choices at a very low wealth (low output and high indebtedness) state. Here, the sovereign would choose to default in equilibrium since the value from defaulting (-2.14) is greater than the maximized value from repaying (-2.58).

Although the maximum value of the objective function is unique, the short-term, long-term portfolio that attains this maximum is not. This is because in such a state, the multiple portfolio combinations yield the same revenue raised, and different portfolios will not affect



$y = 1.0$ $\qquad$ $y = 1.2$

Figure 11: Repayment Heatmap Across Different Output States

the sovereign's continuation value since it is likely to default in the next period anyway. Thus, the outputted optimal policy at this grid point would only be numerical noise, and even the tiniest change in the state space often results in the optimizer choosing a completely different portfolio.

Figure 11 plots the probability of repayment across different indebtedness levels for two different output levels, and shows that the set of points that correspond to repayment more closely resembles a simplex rather than a hypercube, which is what traditional grid-based methods require. Smolyak's method particularly struggles with this issue because it interpolates between points with Chebyshev polynomials which are global, meaning that the policies evaluated at the default points may contaminate the approximation of the policy functions even at points which correspond to repayment.

Users can usse Gaussian process regression to incorporate the irregularity of the domain by simply discarding points which correspond to defaulting in equilibrium when training the Gaussian process for the policy functions. This ensures that the approximation of the policy functions is well-behaved in regions of the state space that matter. Meanwhile, standard grid based methods require all points over the hypercube in order to construct the interpolant.

I adapt my algorithm from Gu and Stangebye (2023).

1. **Initialization**
   (a) I create a level $\mu = 3$ isotropic Smolyak sparse grid corresponding to piecewise linear approximation,[17] which yields 389 points in 6 dimensions. This places grid points at the edge of the hypercube, which mitigates Gaussian Process Regression's tendency to extrapolate poorly. I then add an additional 311 points via Bayesian Active Learning.
   (b) I set my initial guess to correspond to the terminal period of a finite-horizon problem corresponding to a full debt buy-back of all unmatured long-term debt.

[17] The equidistant nodes are allocated according to the Clenshaw-Curtis grid structure.

Set initial guesses:

$$V_T^D(\mathbb{X}_T) = u(h(\mathcal{Y}_T))$$

$$V_T^R(\mathbb{B}, \mathbb{X}) = \max_{\Pi_T} \{u(C_T) - l(\Pi_T - \Pi^*)\}$$

subject to:

$$C_T = \mathcal{Y}_T - \frac{\kappa^S b_T^{SD} + \kappa^L b_T^{LD}}{\Pi_T} - \mathcal{E}_T \left[\kappa_S b_T^{SF} + \kappa_L b_T^{LF}\right] - \overbrace{\mathcal{E}_T q_T^{LF}(1-\delta) b_T^{LF}}^{\text{unmatured LF buyback}} - \overbrace{q_T^{LD} \frac{1-\delta}{\Pi_T} b_T^{LD}}^{\text{unmatured LD buyback}}$$

where

$$q_T^{LF} = \frac{\kappa^L}{1 + 2r + \delta}$$

which is the fixed point to:

$$q^* = \frac{1}{1+r} \times \frac{1}{2} \times \left[\kappa^L + (1-\delta) q^*\right]$$

and $q_T^{LD}(\mathcal{E}_T)$ will be the numerical fixed point:

$$q^*(\mathcal{E}_T) = \frac{1}{1+r} \mathbb{E}_T \left[\frac{1}{2} \times \frac{1}{\bar{\Pi}} \times \frac{\mathcal{E}_T}{\mathcal{E}_{T+1}} \times \left[\kappa^L + (1-\delta) q^*(\mathcal{E}_{T+1})\right]\right]$$

2. **Update**

   (a) Use V_T^R and V_T^D to approximate V_{T-1}^D.

   (b) Use V_T^R, V_T^D, $\hat{\Pi}_T$, q_T^{LF} and q_T^{LD} to approximate $q_{T-1}^{SF}, q_{T-1}^{LF}, q_{T-1}^{SD}, q_{T-1}^{LD}$.

   (c) Use the period $T-1$ pricing functions and V_T^R, V_T^D to approximate V_{T-1}^R and all policy functions.

3. **Convergence Check**

   (a) Evaluate $q_{T-1}^{SF}, q_{T-1}^{LF}, q_{T-1}^{SD}, q_{T-1}^{LD}$ and V_{T-1}^R at a fixed set of 10,000 grid points throughout the state space

   (b) Compute the sup-norm difference.

   (c) If the difference is below the tolerance threshold, stop. Otherwise, set $T = T-1$ and return to step 2 until stationarity.